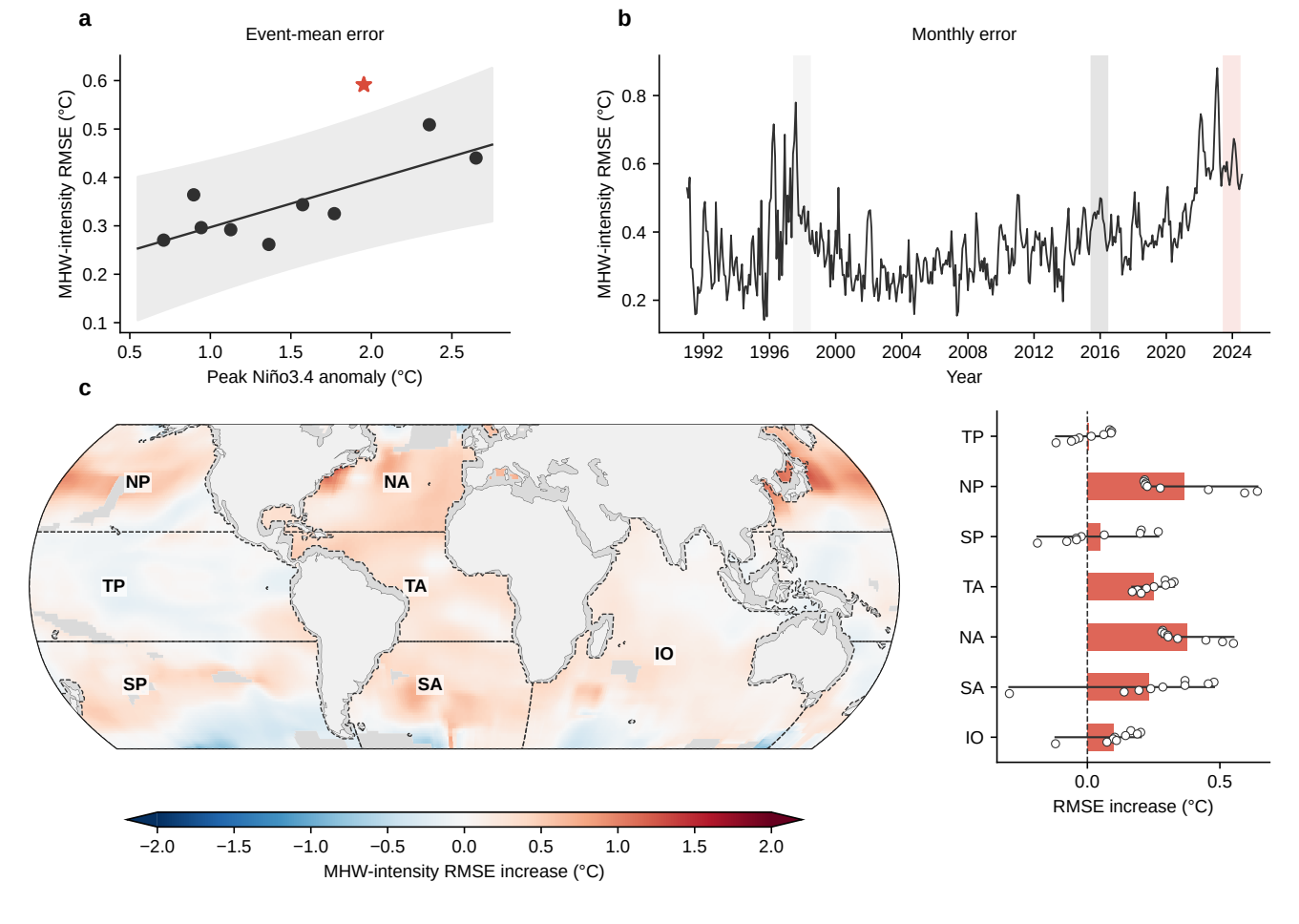
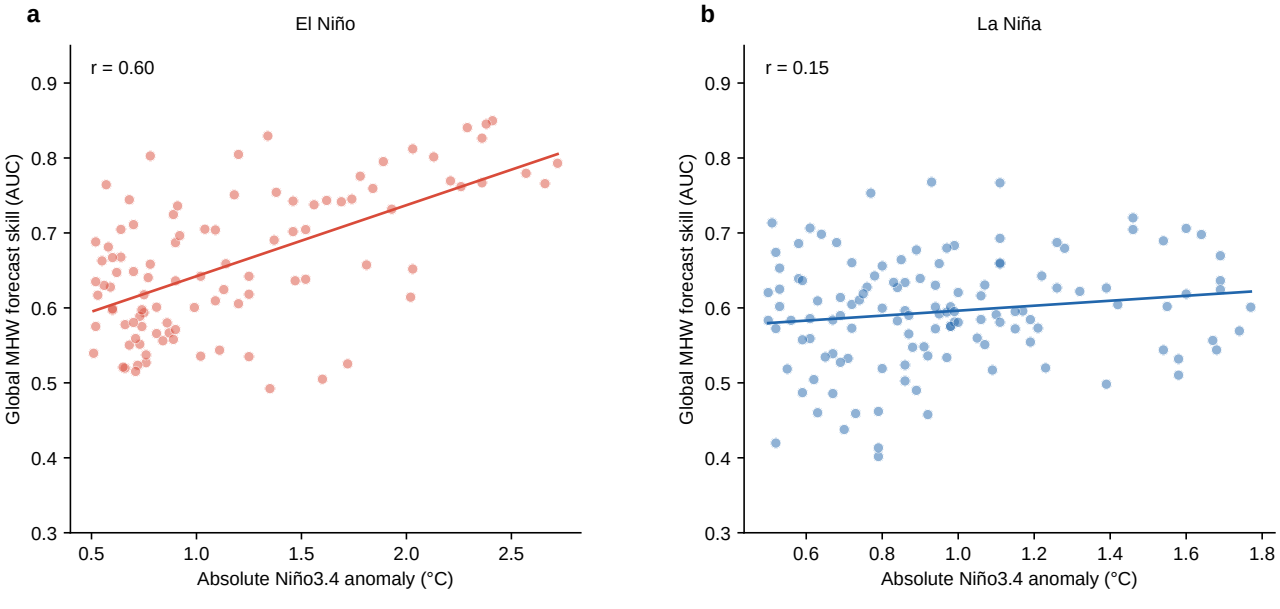
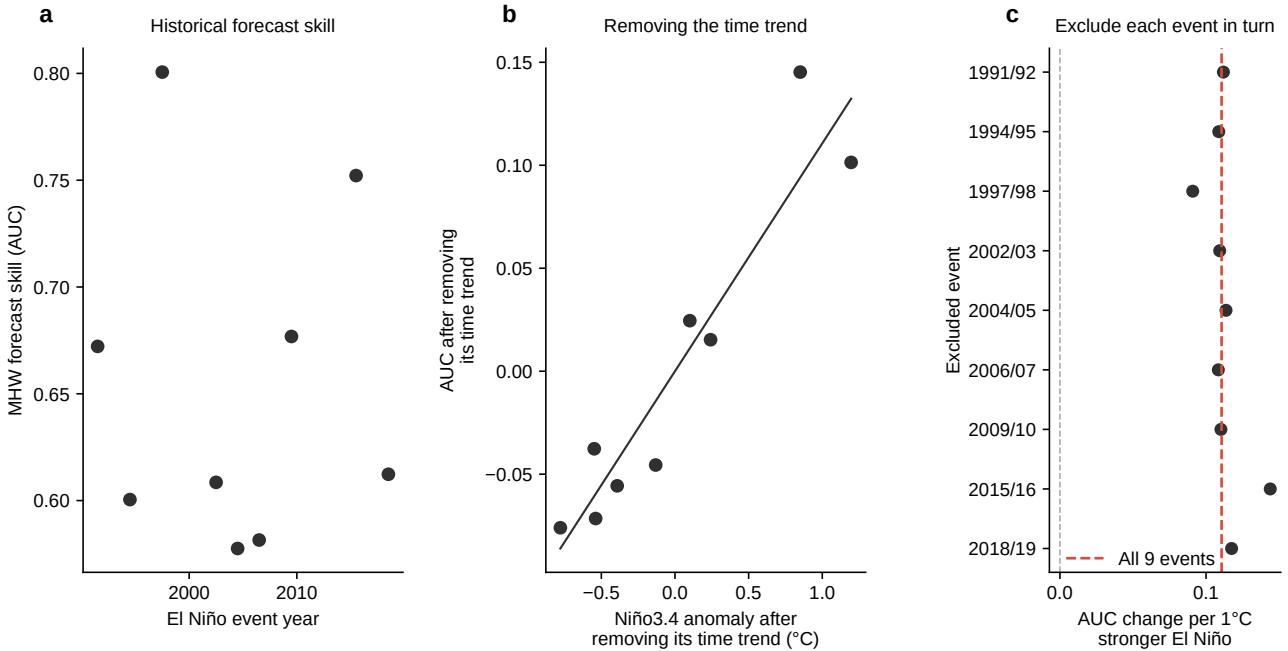
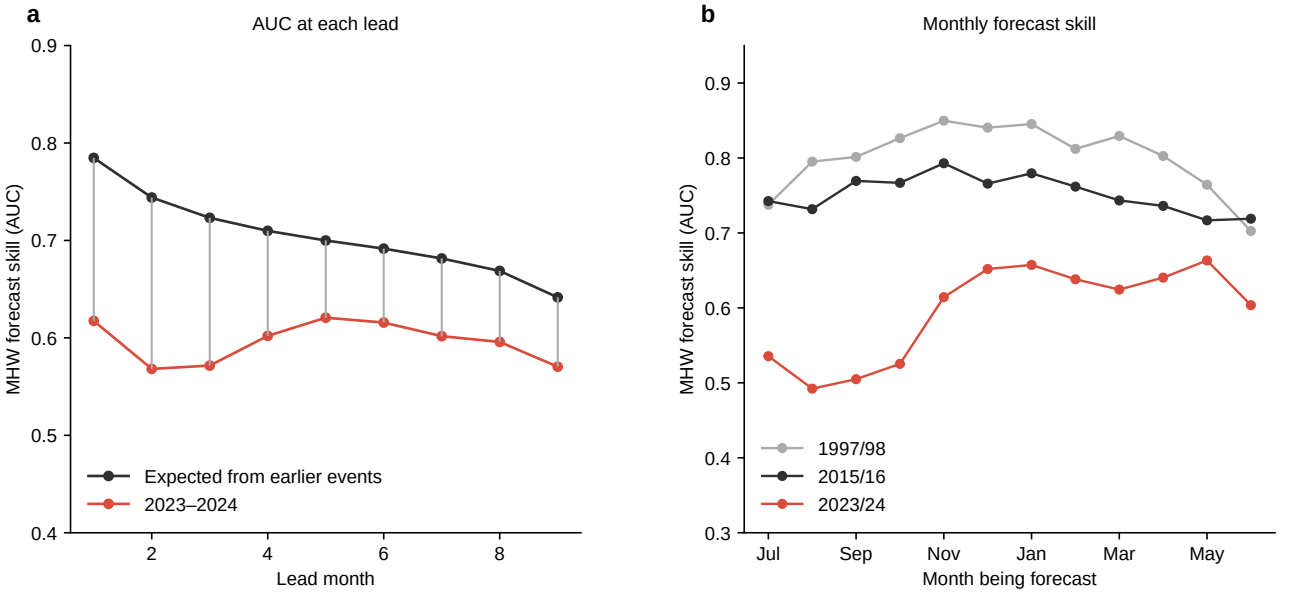
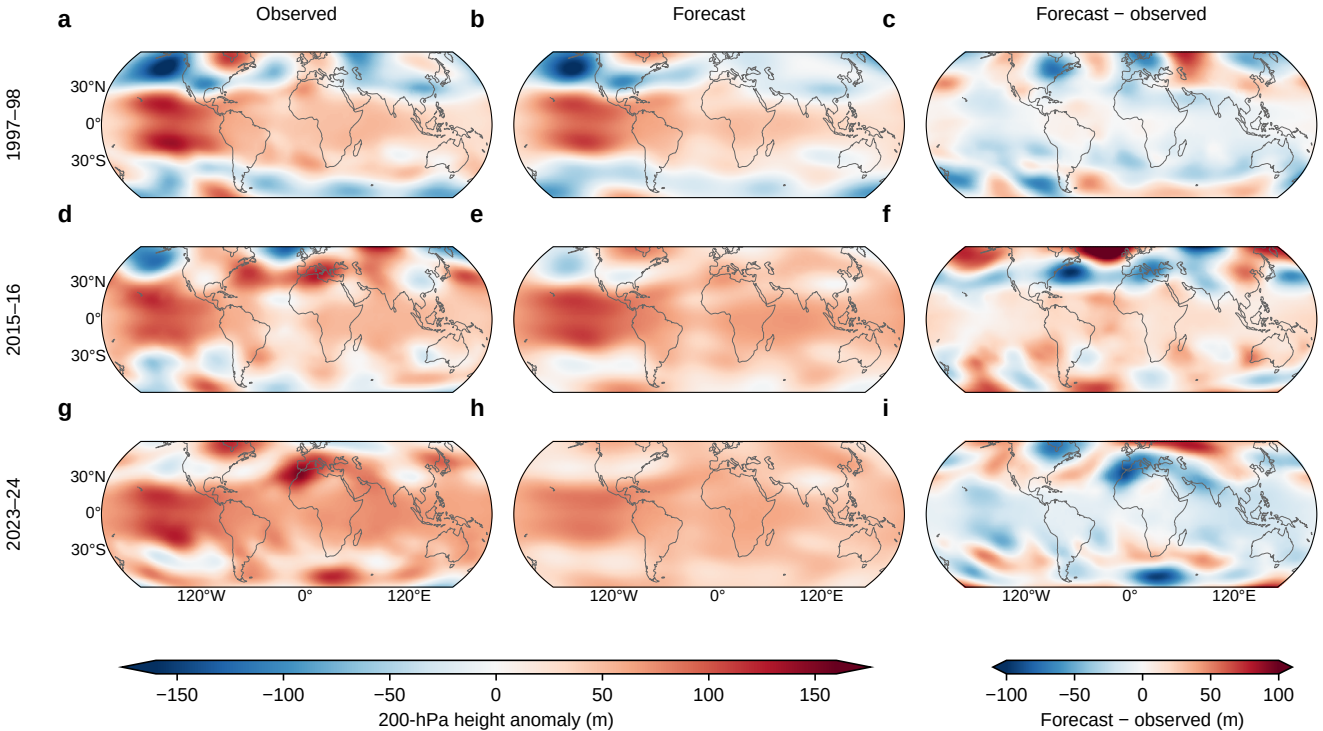

a
Event-mean error
MHW-intensity RMSE (°C)
0.1
0.2
0.3
0.4
0.5
0.6
0.5
1.0
1.5
2.0
2.5
Peak Niño3.4 anomaly (°C)
b
Monthly error
MHW-intensity RMSE (°C)
0.2
0.4
0.6
0.8
1992
1996
2000
2004
2008
2012
2016
2020
2024
Year
c
NP
NA
TP
TA
IO
SP
SA
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
MHW-intensity RMSE increase (°C)
TP
NP
SP
TA
NA
SA
IO
0.0
0.5
RMSE increase (°C)

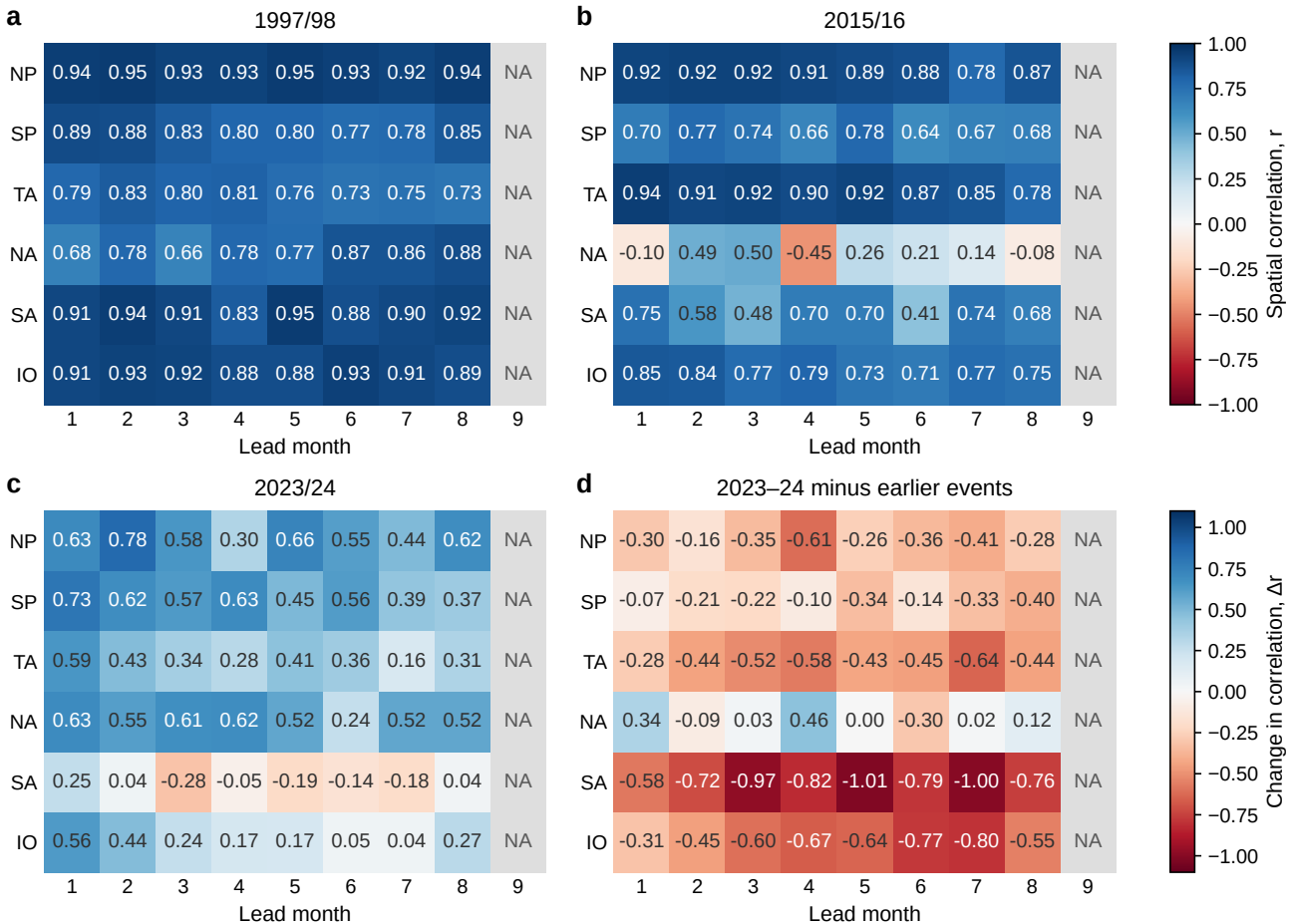

a
El Niño
r = 0.60
Global MHW forecast skill (AUC)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.5
1.0
1.5
2.0
2.5
Absolute Niño3.4 anomaly (°C)
b
La Niña
r = 0.15
Global MHW forecast skill (AUC)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Absolute Niño3.4 anomaly (°C)

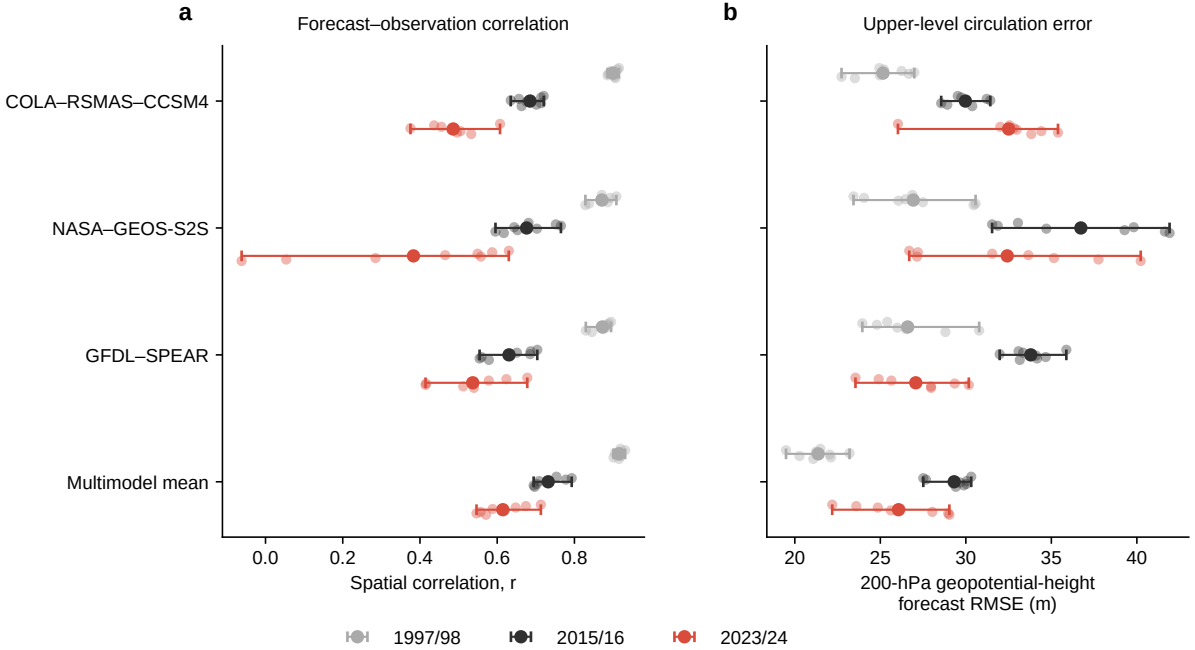

a
Historical forecast skill
MHW forecast skill (AUC)
0.80
0.75
0.70
0.65
0.60
2000
2010
El Niño event year
b
Removing the time trend
AUC after removing its time trend
0.15
0.10
0.05
0.00
−0.05
−0.5
0.0
0.5
1.0
Niño3.4 anomaly after removing its time trend (°C)
c
Exclude each event in turn
Excluded event
1991/92
1994/95
1997/98
2002/03
2004/05
2006/07
2009/10
2015/16
2018/19
All 9 events
0.0
0.1
AUC change per 1°C stronger El Niño

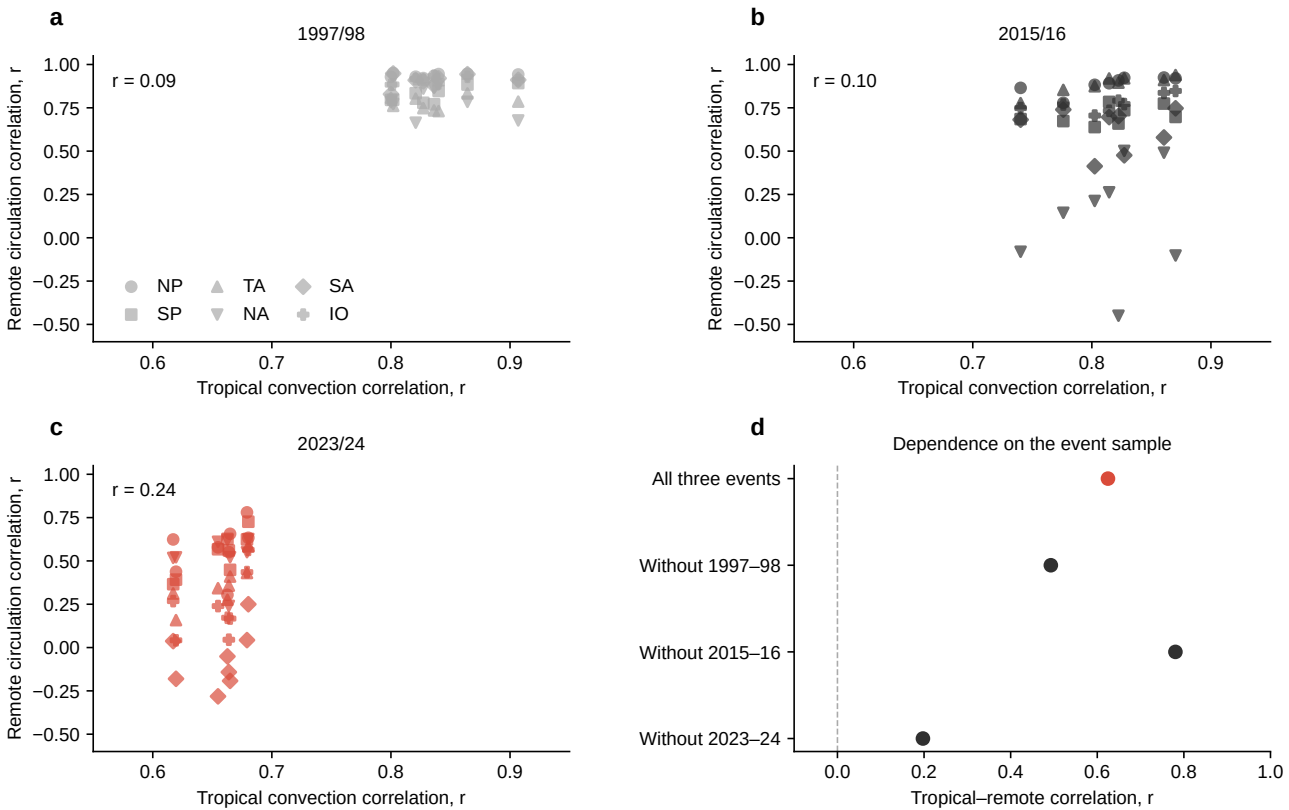

a
AUC at each lead
MHW forecast skill (AUC)
0.9
0.8
0.7
0.6
0.5
0.4
2
4
6
8
Lead month
Expected from earlier events
2023–2024
b
Monthly forecast skill
MHW forecast skill (AUC)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
Jul
Sep
Nov
Jan
Mar
May
Month being forecast
1997/98
2015/16
2023/24

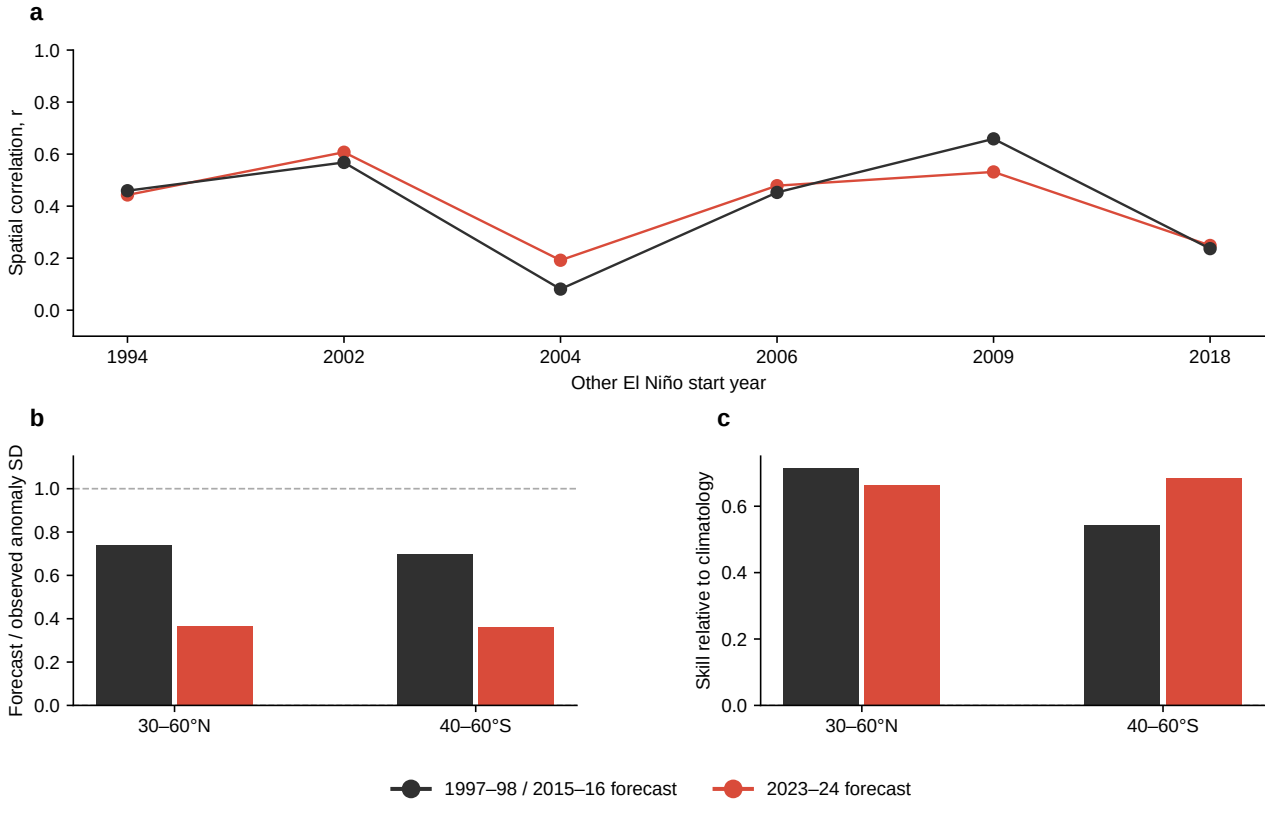
Matched DJF comparison · NASA GEOS-S2S
a
SST
1997–98 0.95 0.94 0.94 0.94 0.94 0.94 0.94 0.94 NA
2015–16 0.94 0.93 0.92 0.91 0.92 0.93 0.93 0.93 NA
2023–24 0.87 0.86 0.85 0.83 0.79 0.63 0.27 0.25 NA
1 2 3 4 5 6 7 8 9
Lead month
b
Correlation decrease (%)
100
50
0
1 3 5 7 9
Lead month
c
Zonal wind stress
1997–98 0.87 0.87 0.84 0.80 0.81 0.85 0.81 0.81 NA
2015–16 0.81 0.74 0.74 0.65 0.70 0.70 0.55 0.69 NA
2023–24 0.40 0.32 0.31 0.31 0.38 0.12 0.13 0.21 NA
1 2 3 4 5 6 7 8 9
Lead month
d
Correlation decrease (%)
100
50
0
1 3 5 7 9
Lead month
e
Convection
1997–98 0.92 0.86 0.78 0.74 0.74 0.83 0.84 0.86 NA
2015–16 0.85 0.82 0.77 0.78 0.84 0.83 0.79 0.72 NA
2023–24 0.69 0.66 0.69 0.68 0.68 0.59 0.41 0.49 NA
1 2 3 4 5 6 7 8 9
Lead month
f
Correlation decrease (%)
100
50
0
1 3 5 7 9
Lead month
−1.0 −0.5 0.0 0.5 1.0
Spatial correlation, r
Historical reference
1997–98
2015–16
Two-event mean

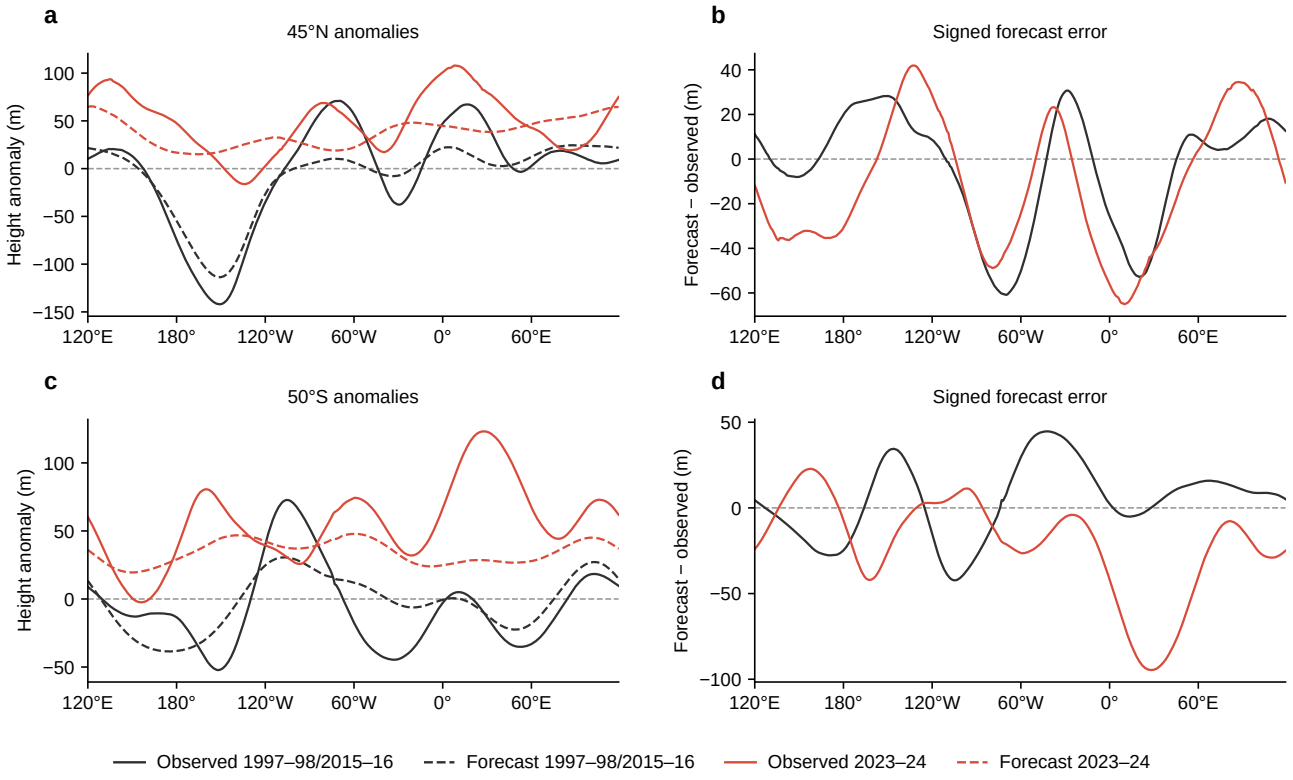

a
Observed
b
Forecast
c
Forecast − observed
1997–98
d
e
f
2015–16
g
h
i
2023–24
30°N
0°
30°S
120°W
0°
120°E
−150
−100
−50
0
50
100
150
200-hPa height anomaly (m)
−100
−50
0
50
100
Forecast − observed (m)

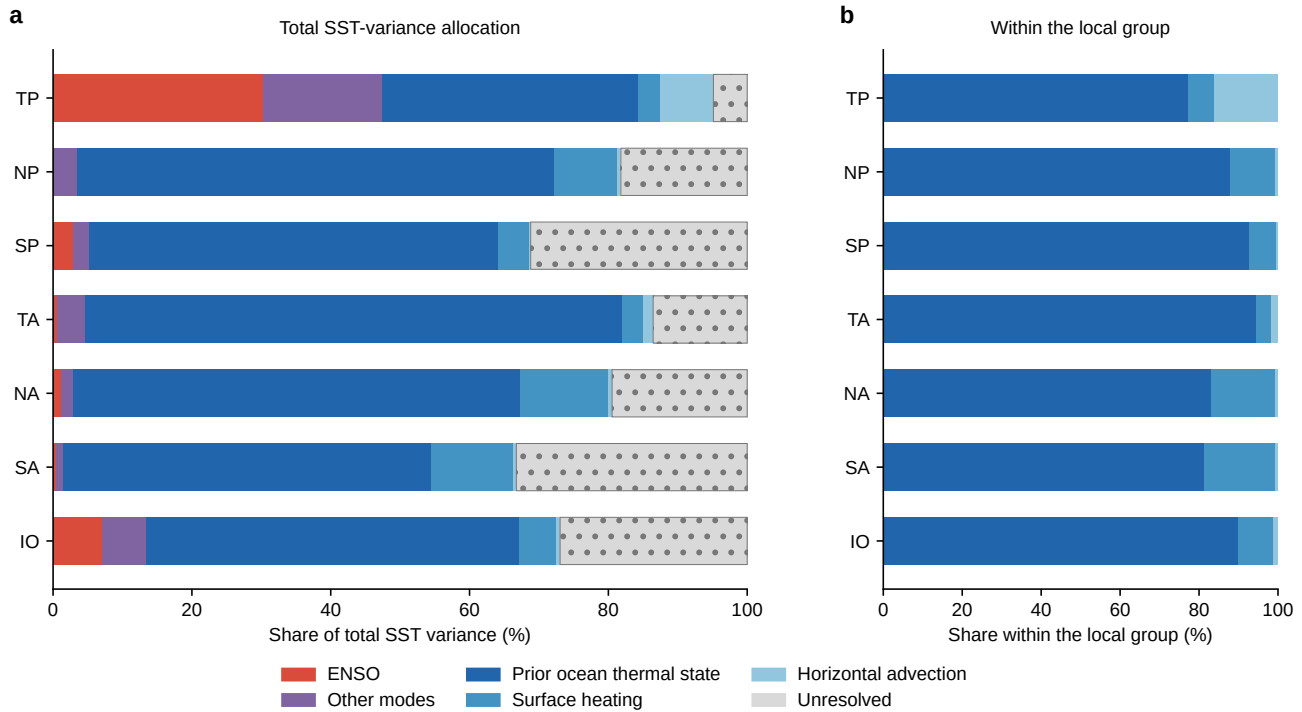

a
1997/98
NP 0.94 0.95 0.93 0.93 0.95 0.93 0.92 0.94 NA
SP 0.89 0.88 0.83 0.80 0.80 0.77 0.78 0.85 NA
TA 0.79 0.83 0.80 0.81 0.76 0.73 0.75 0.73 NA
NA 0.68 0.78 0.66 0.78 0.77 0.87 0.86 0.88 NA
SA 0.91 0.94 0.91 0.83 0.95 0.88 0.90 0.92 NA
IO 0.91 0.93 0.92 0.88 0.88 0.93 0.91 0.89 NA
1 2 3 4 5 6 7 8 9
Lead month
b
2015/16
NP 0.92 0.92 0.92 0.91 0.89 0.88 0.78 0.87 NA
SP 0.70 0.77 0.74 0.66 0.78 0.64 0.67 0.68 NA
TA 0.94 0.91 0.92 0.90 0.92 0.87 0.85 0.78 NA
NA -0.10 0.49 0.50 -0.45 0.26 0.21 0.14 -0.08 NA
SA 0.75 0.58 0.48 0.70 0.70 0.41 0.74 0.68 NA
IO 0.85 0.84 0.77 0.79 0.73 0.71 0.77 0.75 NA
1 2 3 4 5 6 7 8 9
Lead month
1.00 0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75 −1.00
Spatial correlation, r
c
2023/24
NP 0.63 0.78 0.58 0.30 0.66 0.55 0.44 0.62 NA
SP 0.73 0.62 0.57 0.63 0.45 0.56 0.39 0.37 NA
TA 0.59 0.43 0.34 0.28 0.41 0.36 0.16 0.31 NA
NA 0.63 0.55 0.61 0.62 0.52 0.24 0.52 0.52 NA
SA 0.25 0.04 -0.28 -0.05 -0.19 -0.14 -0.18 0.04 NA
IO 0.56 0.44 0.24 0.17 0.17 0.05 0.04 0.27 NA
1 2 3 4 5 6 7 8 9
Lead month
d
2023–24 minus earlier events
NP -0.30 -0.16 -0.35 -0.61 -0.26 -0.36 -0.41 -0.28 NA
SP -0.07 -0.21 -0.22 -0.10 -0.34 -0.14 -0.33 -0.40 NA
TA -0.28 -0.44 -0.52 -0.58 -0.43 -0.45 -0.64 -0.44 NA
NA 0.34 -0.09 0.03 0.46 0.00 -0.30 0.02 0.12 NA
SA -0.58 -0.72 -0.97 -0.82 -1.01 -0.79 -1.00 -0.76 NA
IO -0.31 -0.45 -0.60 -0.67 -0.64 -0.77 -0.80 -0.55 NA
1 2 3 4 5 6 7 8 9
Lead month
1.00 0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75 −1.00
Change in correlation, Δr

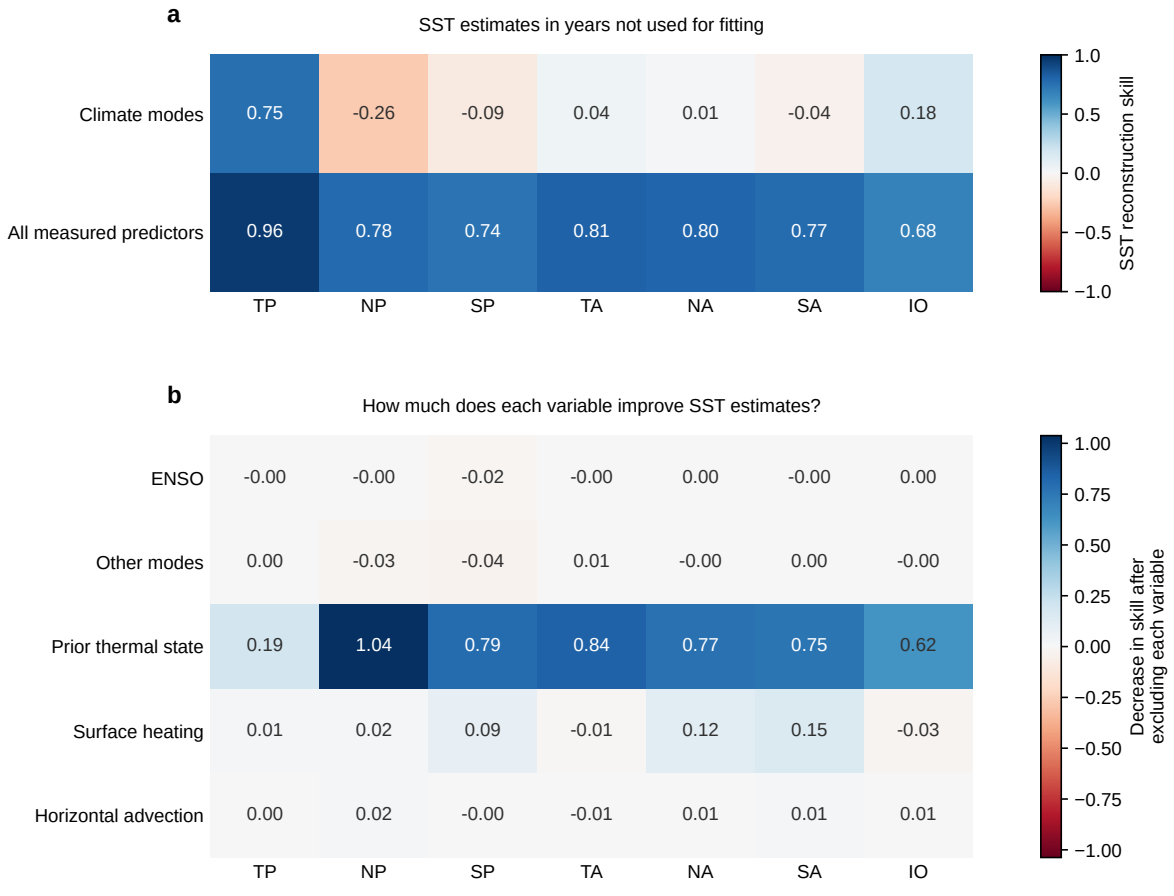

a
Forecast–observation correlation
COLA–RSMAS–CCSM4
NASA–GEOS-S2S
GFDL–SPEAR
Multimodel mean
0.0
0.2
0.4
0.6
0.8
Spatial correlation, r
b
Upper-level circulation error
20
25
30
35
40
200-hPa geopotential-height forecast RMSE (m)
1997/98
2015/16
2023/24

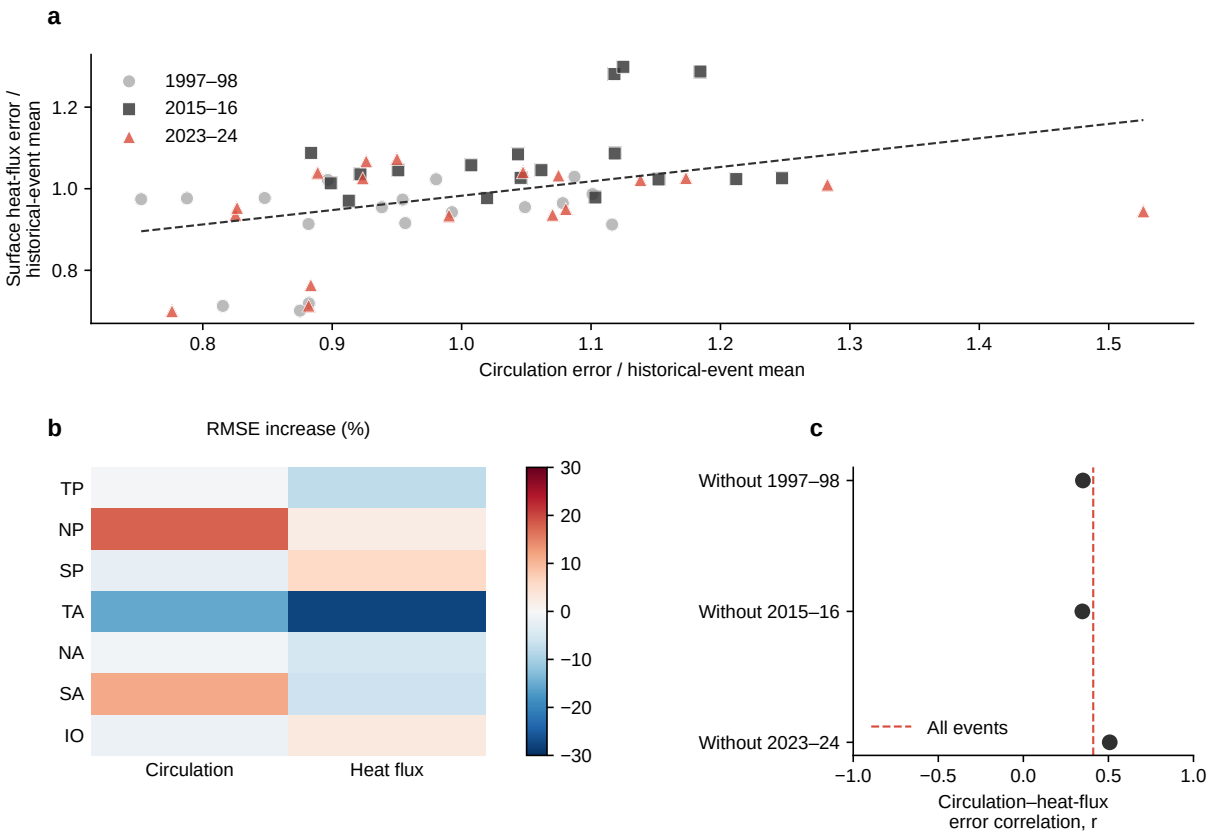

a
1997/98
r = 0.09
Remote circulation correlation, r
Tropical convection correlation, r
NP
SP
TA
NA
SA
IO
b
2015/16
r = 0.10
c
2023/24
r = 0.24
d
Dependence on the event sample
All three events
Without 1997–98
Without 2015–16
Without 2023–24
Tropical–remote correlation, r

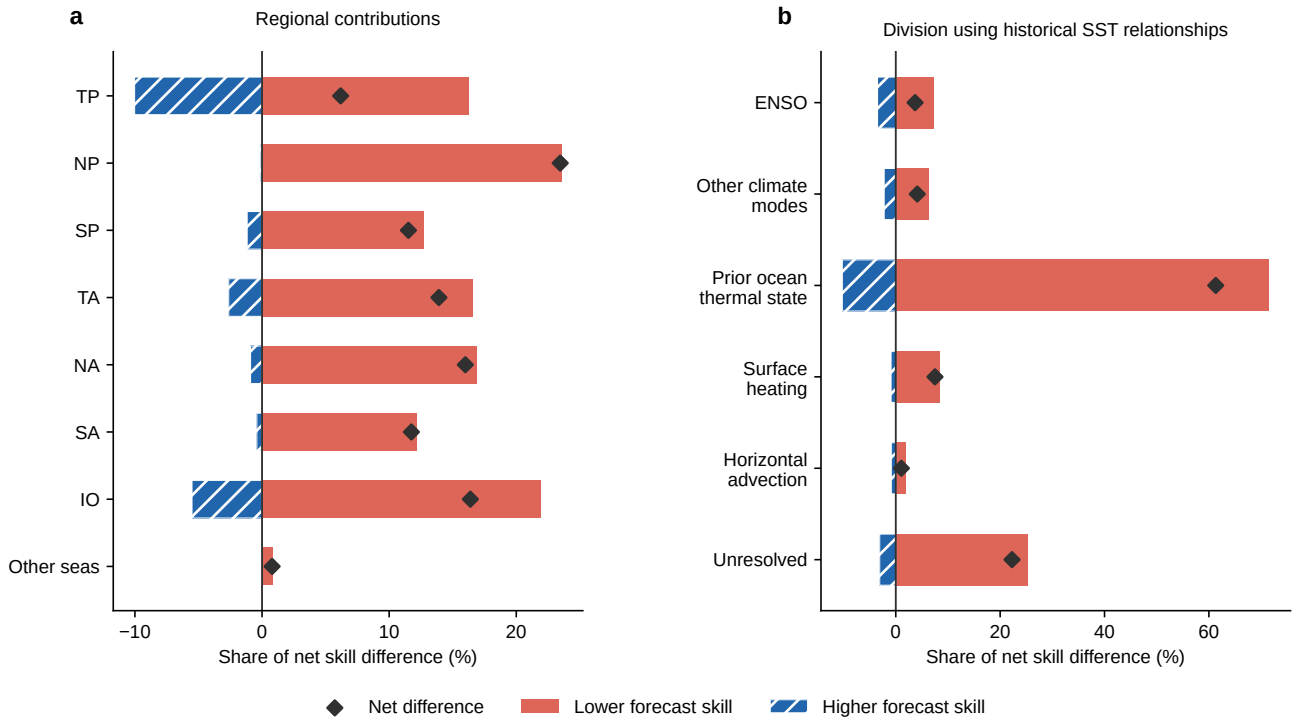

a
1.0
0.8
0.6
0.4
0.2
0.0
Spatial correlation, r
1994
2002
2004
2006
2009
2018
Other El Niño start year
b
Forecast / observed anomaly SD
1.0
0.8
0.6
0.4
0.2
0.0
30–60°N
40–60°S
c
Skill relative to climatology
0.6
0.4
0.2
0.0
30–60°N
40–60°S
1997–98 / 2015–16 forecast
2023–24 forecast

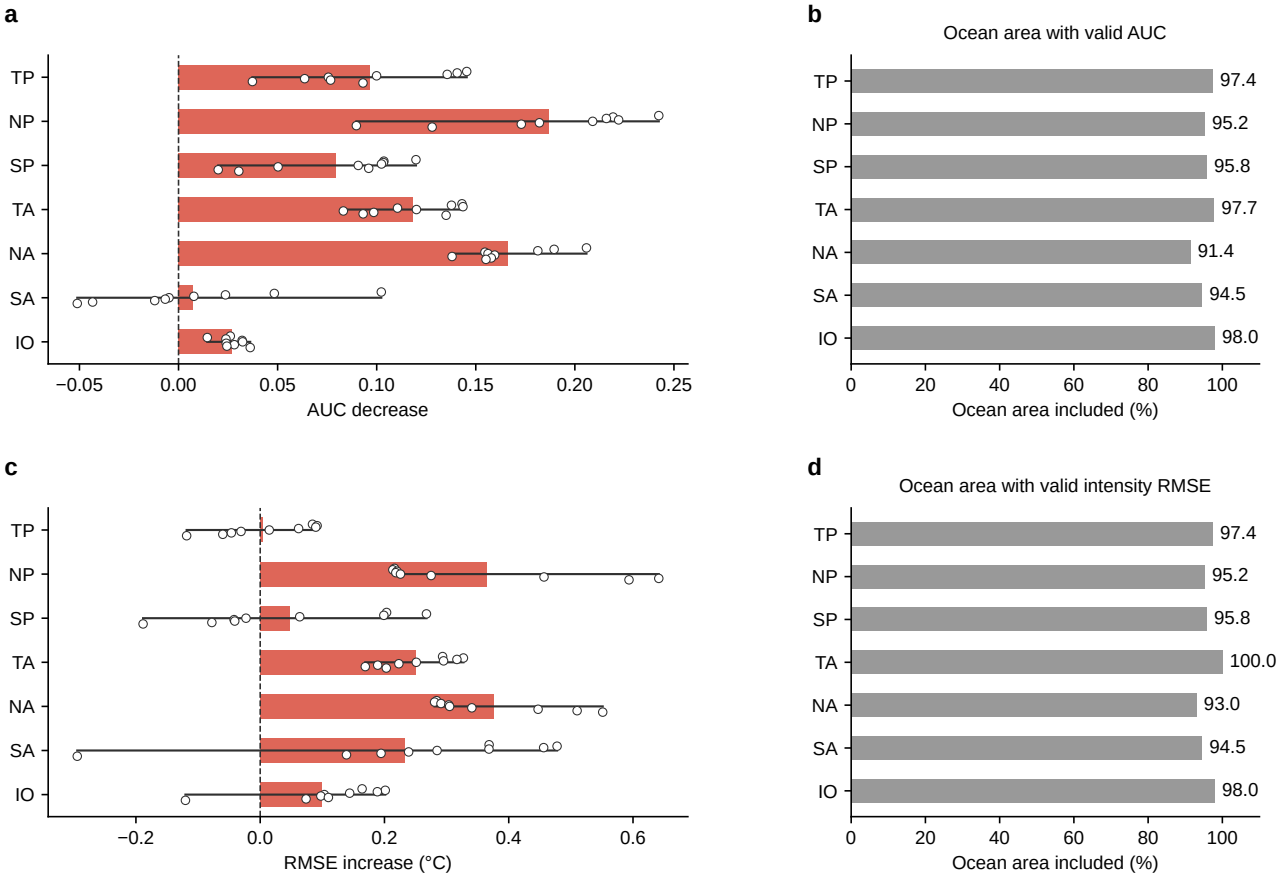

a
45°N anomalies
Height anomaly (m)
100
50
0
−50
−100
−150
120°E
180°
120°W
60°W
0°
60°E
b
Signed forecast error
Forecast − observed (m)
40
20
0
−20
−40
−60
120°E
180°
120°W
60°W
0°
60°E
c
50°S anomalies
Height anomaly (m)
100
50
0
−50
120°E
180°
120°W
60°W
0°
60°E
d
Signed forecast error
Forecast − observed (m)
50
0
−50
−100
120°E
180°
120°W
60°W
0°
60°E
Observed 1997–98/2015–16
Forecast 1997–98/2015–16
Observed 2023–24
Forecast 2023–24

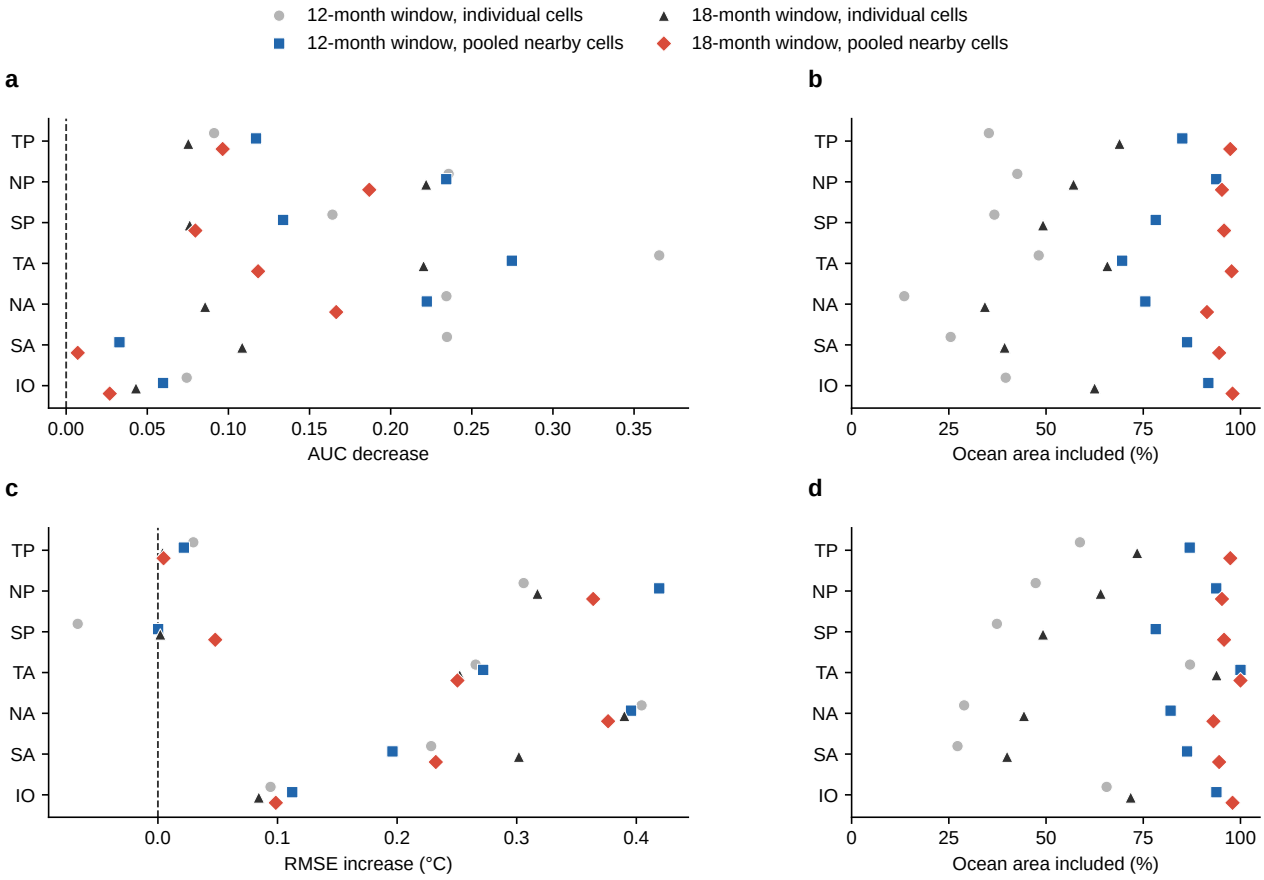

a
Total SST-variance allocation
TP
NP
SP
TA
NA
SA
IO
0
20
40
60
80
100
Share of total SST variance (%)
b
Within the local group
TP
NP
SP
TA
NA
SA
IO
0
20
40
60
80
100
Share within the local group (%)
ENSO
Other modes
Prior ocean thermal state
Surface heating
Horizontal advection
Unresolved


a
SST estimates in years not used for fitting
Climate modes
All measured predictors
0.75
-0.26
-0.09
0.04
0.01
-0.04
0.18
0.96
0.78
0.74
0.81
0.80
0.77
0.68
TP
NP
SP
TA
NA
SA
IO
1.0
0.5
0.0
−0.5
−1.0
SST reconstruction skill
b
How much does each variable improve SST estimates?
ENSO
Other modes
Prior thermal state
Surface heating
Horizontal advection
-0.00 -0.00 -0.02 -0.00 0.00 -0.00 0.00
0.00 -0.03 -0.04 0.01 -0.00 0.00 -0.00
0.19 1.04 0.79 0.84 0.77 0.75 0.62
0.01 0.02 0.09 -0.01 0.12 0.15 -0.03
0.00 0.02 -0.00 -0.01 0.01 0.01 0.01
TP
NP
SP
TA
NA
SA
IO
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Decrease in skill after excluding each variable


a
1997–98
2015–16
2023–24
Surface heat-flux error / historical-event mean
1.2
1.0
0.8
0.8
0.9
1.0
1.1
1.2
1.3
1.4
1.5
Circulation error / historical-event mean
b
RMSE increase (%)
TP
NP
SP
TA
NA
SA
IO
Circulation
Heat flux
30
20
10
0
−10
−20
−30
c
Without 1997–98
Without 2015–16
Without 2023–24
All events
−1.0
−0.5
0.0
0.5
1.0
Circulation–heat-flux error correlation, r


a
Regional contributions
TP
NP
SP
TA
NA
SA
IO
Other seas
−10
0
10
20
Share of net skill difference (%)
b
Division using historical SST relationships
ENSO
Other climate modes
Prior ocean thermal state
Surface heating
Horizontal advection
Unresolved
0
20
40
60
Share of net skill difference (%)
Net difference
Lower forecast skill
Higher forecast skill


a
TP
NP
SP
TA
NA
SA
IO
−0.05
0.00
0.05
0.10
0.15
0.20
0.25
AUC decrease
b
Ocean area with valid AUC
97.4
95.2
95.8
97.7
91.4
94.5
98.0
0
20
40
60
80
100
Ocean area included (%)
c
−0.2
0.0
0.2
0.4
0.6
RMSE increase (°C)
d
Ocean area with valid intensity RMSE
100.0
93.0


12-month window, individual cells
12-month window, pooled nearby cells
18-month window, individual cells
18-month window, pooled nearby cells
a
b
c
d
TP
NP
SP
TA
NA
SA
IO
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
AUC decrease
0
25
50
75
100
Ocean area included (%)
0.0
0.1
0.2
0.3
0.4
RMSE increase (°C)